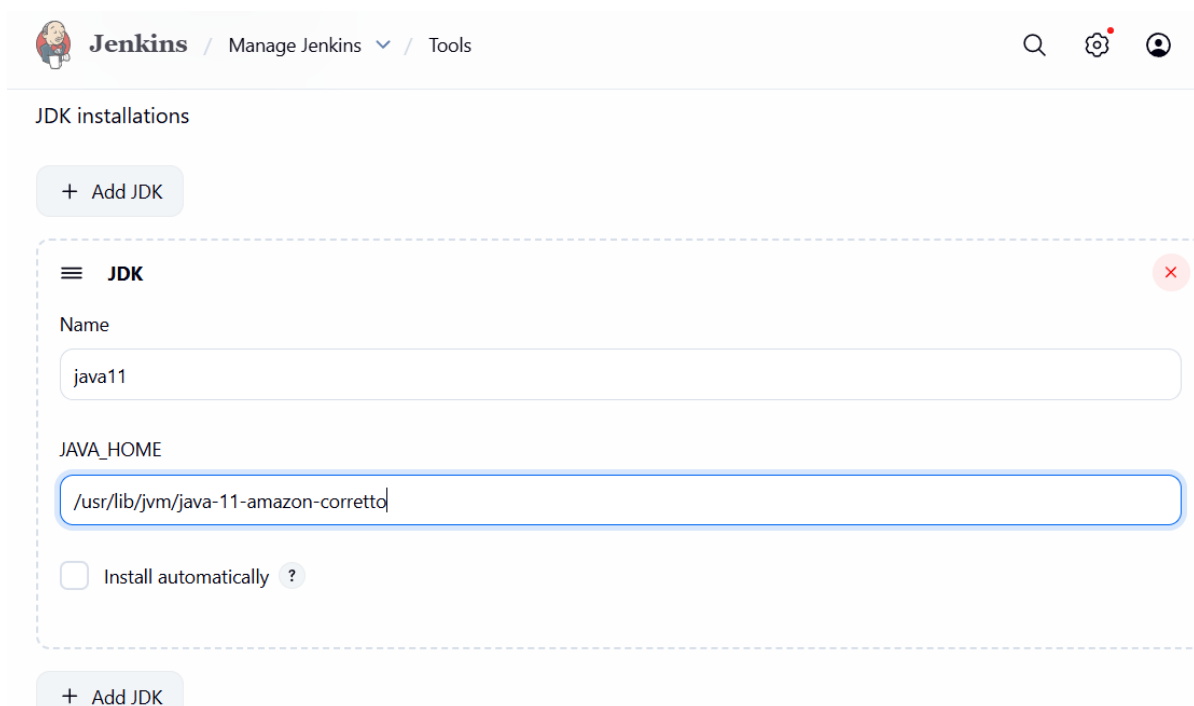


02 Assignment - Jenkins

Name: Pooja Prakash Belgaumkar

1. L2 – Create CICD Pipeline to Clone and Build Java Maven Web Application.

- Install the JDK and Maven to complete the CICD pipeline tasks effectively.



Jenkins / Manage Jenkins / Tools

JDK installations

+ Add JDK

JDK

Name

java11

JAVA_HOME

/usr/lib/jvm/java-11-amazon-corretto

☐ Install automatically ?

+ Add JDK

 Jenkins

Manage Jenkins

Tools

Search

Settings

User

Maven installations

+ Add Maven

≡ Maven

Name

maven3

☒ Install automatically

?

≡ Install from Apache

Version

3.9.12

+ Add Installer

+ Add Maven

Save

Apply

- Create the Jenkins pipeline job named java-maven-cicd.

 Jenkins

All

New Item

Search

Settings

User

New Item

Enter an item name

java-maven-cicd

Select an item type

 Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 Pipeline
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

 Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

 Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK

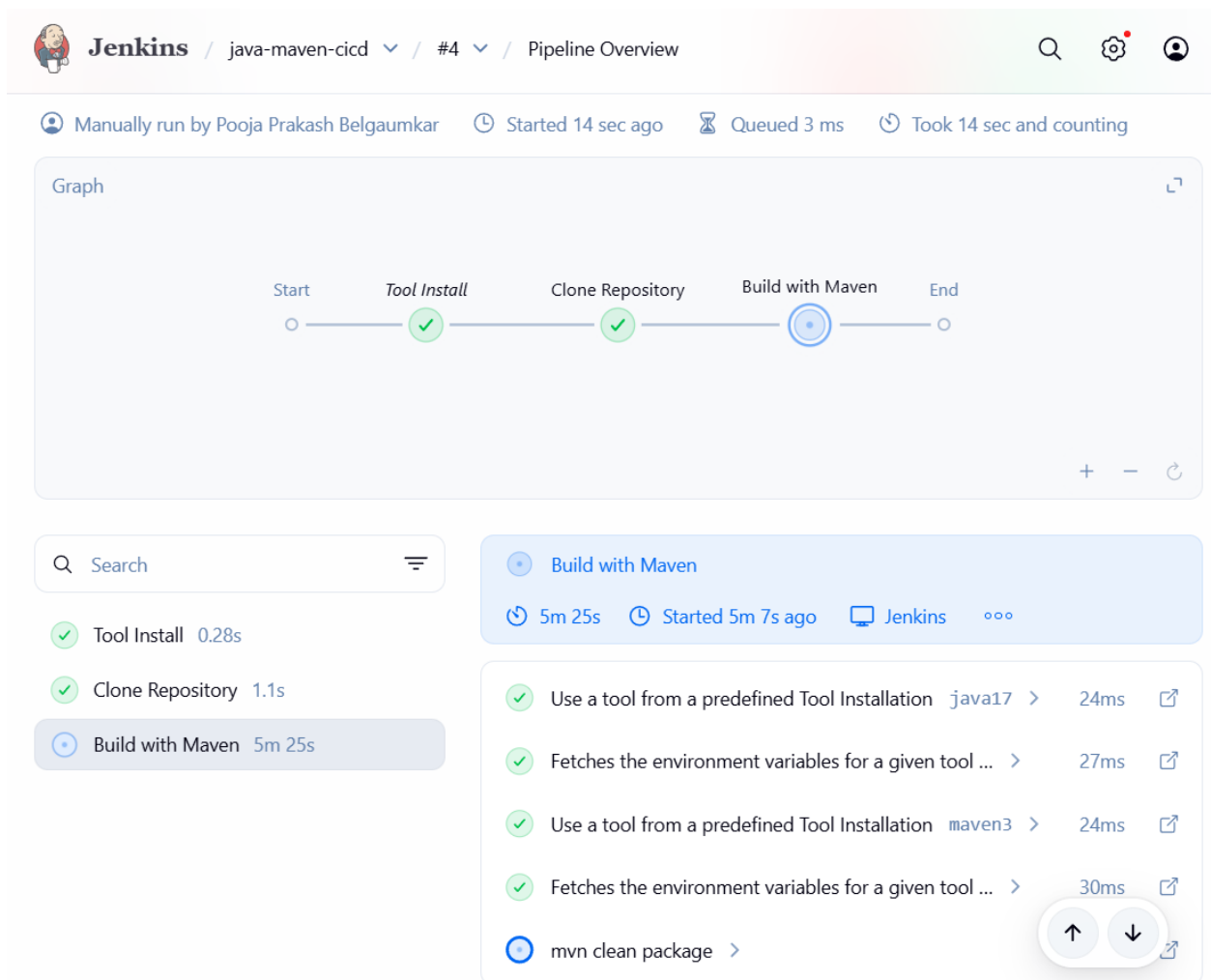
- Add the pipeline script, each stage has logical step.
- Checkout Stage: This stage downloads the latest project code from the Git repository into Jenkins.
- Build Stage: This stage compiles the Java project using Maven to check for build errors.
- Test Stage: This stage runs test cases to verify that the application works correctly.
- Archive Stage: This stage saves the generated build file for future use or deployment.

The screenshot shows the Jenkins web interface for configuring a pipeline. The breadcrumb navigation at the top reads 'Jenkins / java-maven-cicd / Configuration'. On the left, the 'Configure' section has four tabs: 'General', 'Triggers', 'Pipeline' (which is selected and highlighted), and 'Advanced'. The main content area is titled 'Pipeline' and includes the instruction 'Define your Pipeline using Groovy directly or pull it from source control.' Below this, the 'Definition' dropdown is set to 'Pipeline script'. A 'Script' section with a help icon contains a text area with the following Groovy pipeline script:

```
1 pipeline {
2     agent any
3
4     tools {
5         jdk 'java11'
6         maven 'maven3'
7     }
8
9     stages {
10
11         stage('Clone Repository') {
12             steps {
13                 git branch: 'main',
14                     url: 'https://github.com/spring-pr
15
```

At the bottom of the script area, there is a checkbox labeled 'Use Groovy Sandbox' which is checked. Below the script area are two buttons: 'Save' and 'Apply'.

- Then click save and Build Now, we will see console output logs and each stage executing.



- In this task, Jenkins is used to automatically get the Java Maven project from GitHub and build it. This helps in checking whether the code works correctly every time changes are made.